

Assignment Brief: AI-Powered Contact Center Analytics

Role: Senior Consultant – AI/ML

Estimated Duration: 5–8 hours

Submission Format: GitHub repo or zipped folder + optional dashboard demo

Objective

Design an AI/ML-driven analytics system that processes contact center data to extract actionable insights aimed at improving customer experience and operational efficiency.

Key Deliverables

1. Data Preprocessing

- Clean and normalize call logs (e.g., durations, agent IDs, IVR paths). IVR Path or flow is a checkpoint logged after every step to show the progression. If data is not there, you are encouraged to make up your own data.
- Handle missing values and standardize columns.
- Anonymize sensitive information for privacy compliance.

2. Exploratory Data Analysis (EDA)

- Identify the **most common IVR/customer paths** and call flow drop-offs.
- Analyze **Agent Handling Time** and **First Call Resolution (FCR)** rates.
- Map and analyze **transfer patterns** from self-service bots to live agents.
- Extract **common intents** or topics from customer utterances.

3. Sentiment Analysis (you can make up your own transcripts simulating agent customer conversation or download them from anywhere)

- Perform NLP sentiment analysis across the call transcript.
- Show **how sentiment changes throughout the call lifecycle**.
- Identify sentiment trends by team, agent, topic, or call resolution.

4. Agent & Supervisor Metrics (hierarchy data if not available, you are encouraged to create your own made-up data)

- Derive individual agent KPIs: AHT, FCR, Escalations, Sentiment Score.
- Aggregate metrics **by team/supervisor** based on org hierarchy.

- Show performance variability and trends by time or customer segment.

5. Bot Performance Evaluation (If data not present, you can use made-up data)

- Identify and analyze:
 - **No Match** responses
 - **No Input** events
 - **Live agent transfers** by reason
- Recommend how to improve self-service recognition and routing.

6. Data Visualization

- Build clear, insightful charts or dashboards using tools like:
 - Plotly, Dash, Streamlit, Power BI, or Tableau
- Add **filtering by agent, date range, topic, or team**

7. Evaluation Strategy

- Justify the metrics chosen (e.g., F1 score, AUC for models; accuracy for bot classification).
- Discuss how you would evaluate **LLM outputs** if used (e.g., hallucination checks, relevance).

8. Summary Report

- Write a 1-2 page summary or 4–5 slide deck with:
 - Key insights
 - Methodology
 - Decisions taken & why
 - Recommendations

Evaluation Criteria

Category

What We're Looking For

Technical Depth Use of relevant ML/NLP/analytics techniques

Decision-Making Justified choices in data handling, model use, metrics

Communication Clear explanation, story, and visualizations

Category	What We're Looking For
Innovation	Extra effort, unique metrics, or actionable ideas

Submission Instructions

Please submit within **5 days** of receiving this assignment.

 Include:

- GitHub repo or ZIP file with:
 - Source code and documentation
 - README or slide deck summary
 - Sample visualizations or exported dashboards
 - Optional: Hosted Streamlit/Dash demo link
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Suggested Public Datasets

- [Kaggle – Call Center Data](#)
- [Kaggle – PWC Call Centre Dataset](#)

These Datasets don't have all the data you require for doing the things I am asking you to do in this assignment. What is not there, you are welcome to research and create your own made-up data. Meaning - you may supplement this with synthetic or sample data to simulate richer call logs, IVR flows, or chatbot outputs.